

CS 3793: Assignment-3 Output

Stochastic Gradient Descent

Accuracy for LR of 0.05 on Training data = 0.68

Accuracy for LR of 0.05 on Testing data = 0.7225

Accuracy for LR of 0.1 on Training data = 0.68

Accuracy for LR of 0.1 on Testing data = 0.7175

Accuracy for LR of 0.15 on Training data = 0.69

Accuracy for LR of 0.15 on Testing data = 0.7175

Accuracy for LR of 0.2 on Training data = 0.69

Accuracy for LR of 0.2 on Testing data = 0.715

Accuracy for LR of 0.25 on Training data = 0.69

Accuracy for LR of 0.25 on Testing data = 0.71

Accuracy for LR of 0.3 on Training data = 0.69

Accuracy for LR of 0.3 on Testing data = 0.7075

Accuracy for LR of 0.35 on Training data = 0.7

Accuracy for LR of 0.35 on Testing data = 0.7

Accuracy for LR of 0.4 on Training data = 0.7

Accuracy for LR of 0.4 on Testing data = 0.69

Accuracy for LR of 0.45 on Training data = 0.69

Accuracy for LR of 0.45 on Testing data = 0.685

Accuracy for LR of 0.5 on Training data = 0.69

Accuracy for LR of 0.5 on Testing data = 0.6825

Accuracy for LR of 0.55 on Training data = 0.69

Accuracy for LR of 0.55 on Testing data = 0.675

Accuracy for LR of 0.6 on Training data = 0.69

Accuracy for LR of 0.6 on Testing data = 0.6625

Accuracy for LR of 0.65 on Training data = 0.69

Accuracy for LR of 0.65 on Testing data = 0.6575

Accuracy for LR of 0.7 on Training data = 0.69

Accuracy for LR of 0.7 on Testing data = 0.65

Accuracy for LR of 0.75 on Training data = 0.68

Accuracy for LR of 0.75 on Testing data = 0.65

Accuracy for LR of 0.8 on Training data = 0.67

Accuracy for LR of 0.8 on Testing data = 0.65

Accuracy for LR of 0.85 on Training data = 0.66

Accuracy for LR of 0.85 on Testing data = 0.645

Accuracy for LR of 0.9 on Training data = 0.66

Accuracy for LR of 0.9 on Testing data = 0.6425

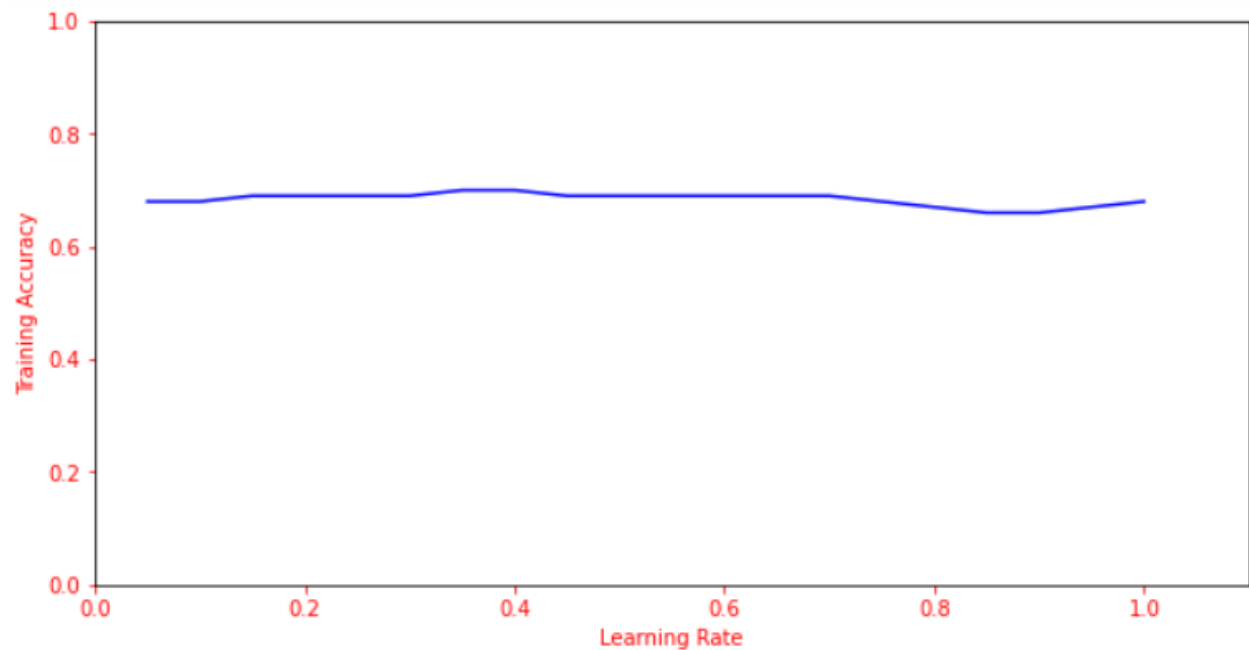
Accuracy for LR of 0.95 on Training data = 0.67

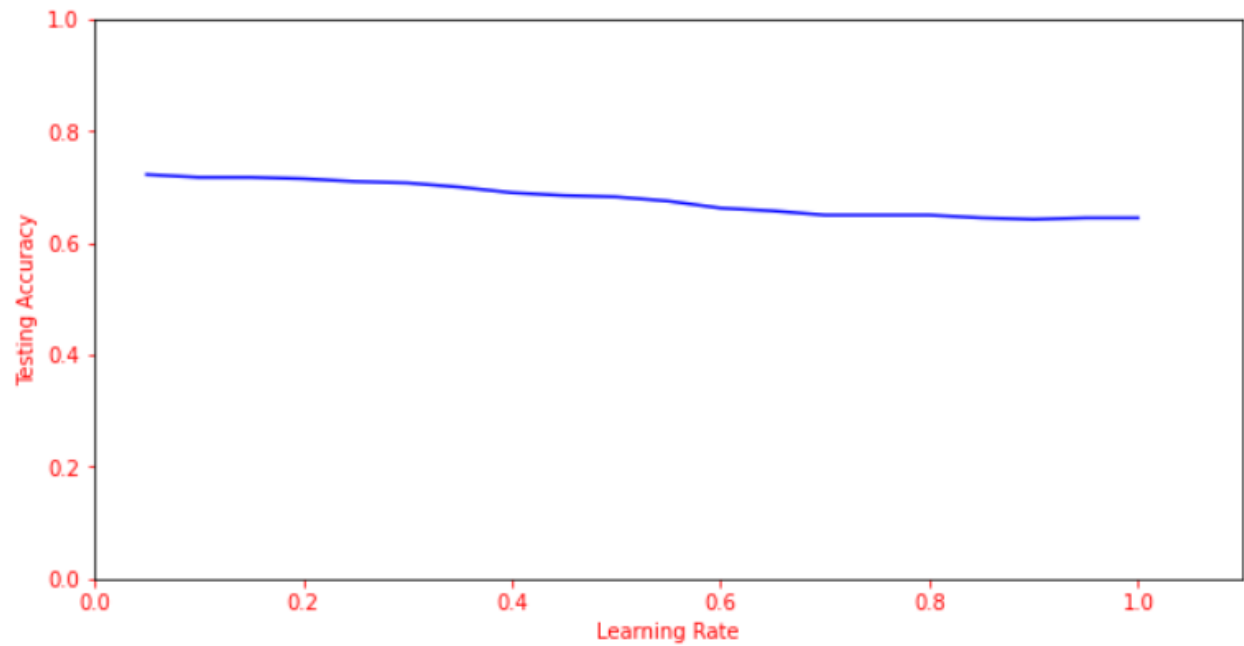
Accuracy for LR of 0.95 on Testing data = 0.645

Accuracy for LR of 1.0 on Training data = 0.68

Accuracy for LR of 1.0 on Testing data = 0.645

Extra Credit - Accuracy Plots





ID3 Decision Tree

attr5 = 0 :

attr6 = 0 :

attr2 = 0 :

attr1 = 0 :

attr4 = 0 :

attr3 = 0 : 0.0 -- 9

attr3 = 1 : 0.0 -- 12

attr4 = 1 :

attr3 = 0 : 0.0 -- 13

attr3 = 1 : 0.0 -- 14

attr1 = 1 :

attr4 = 0 :

attr3 = 0 : 0.0 -- 15

attr3 = 1 : 0.0 -- 10

```
attr4 = 1 : 0.0 -- 30
attr2 = 1 :
  attr4 = 0 :
    attr3 = 0 : 0.0 -- 25
    attr3 = 1 :
      attr1 = 0 : 0.0 -- 11
      attr1 = 1 : 0.0 -- 18
  attr4 = 1 :
    attr1 = 0 :
      attr3 = 0 : 0.0 -- 10
      attr3 = 1 : 0.0 -- 18
    attr1 = 1 :
      attr3 = 0 : 0.0 -- 12
      attr3 = 1 : 0.0 -- 17
attr6 = 1 :
  attr4 = 0 :
    attr2 = 0 :
      attr3 = 0 :
        attr1 = 0 : 0.0 -- 11
        attr1 = 1 : 0.0 -- 13
      attr3 = 1 :
        attr1 = 0 : 0.0 -- 9
        attr1 = 1 : 0.0 -- 9
    attr2 = 1 :
      attr1 = 0 :
        attr3 = 0 : 0.0 -- 10
        attr3 = 1 : 0.0 -- 11
      attr1 = 1 :
        attr3 = 0 : 0.0 -- 14
```

```
attr3 = 1 : 0.0 -- 16
attr4 = 1 :
  attr2 = 0 :
    attr1 = 0 :
      attr3 = 0 : 0.0 -- 18
      attr3 = 1 : 0.0 -- 12
    attr1 = 1 :
      attr3 = 0 : 0.0 -- 13
      attr3 = 1 : 0.0 -- 10
  attr2 = 1 :
    attr1 = 0 :
      attr3 = 0 : 1.0 -- 11
      attr3 = 1 : 1.0 -- 9
    attr1 = 1 :
      attr3 = 0 : 0.0 -- 9
      attr3 = 1 : 0 -- 8
attr5 = 1 :
  attr6 = 0 :
    attr3 = 0 :
      attr4 = 0 :
        attr1 = 0 :
          attr2 = 0 : 0.0 -- 14
          attr2 = 1 : 0.0 -- 11
        attr1 = 1 :
          attr2 = 0 : 0.0 -- 8
          attr2 = 1 : 0.0 -- 14
      attr4 = 1 :
        attr1 = 0 :
          attr2 = 0 : 1.0 -- 16
```

```
attr2 = 1 : 1.0 -- 20
attr1 = 1 :
attr2 = 0 : 0.0 -- 11
attr2 = 1 : 0.0 -- 9
attr3 = 1 :
attr2 = 0 :
attr4 = 0 :
attr1 = 0 : 1.0 -- 13
attr1 = 1 : 1.0 -- 14
attr4 = 1 : 1.0 -- 28
attr2 = 1 :
attr1 = 0 :
attr4 = 0 : 1.0 -- 14
attr4 = 1 : 1.0 -- 9
attr1 = 1 :
attr4 = 0 : 1.0 -- 8
attr4 = 1 : 0.0 -- 14
attr6 = 1 :
attr4 = 0 :
attr2 = 0 :
attr3 = 0 :
attr1 = 0 : 0.0 -- 18
attr1 = 1 : 0.0 -- 11
attr3 = 1 : 0.0 -- 26
attr2 = 1 :
attr1 = 0 :
attr3 = 0 : 0.0 -- 14
attr3 = 1 : 0.0 -- 14
attr1 = 1 : 0.0 -- 14
```

```

attr4 = 1 :
    attr1 = 0 :
        attr2 = 0 :
            attr3 = 0 : 0.0  -- 11
            attr3 = 1 : 1.0  -- 13
        attr2 = 1 :
            attr3 = 0 : 1.0  -- 12
            attr3 = 1 : 0.0  -- 14
    attr1 = 1 :
        attr2 = 0 :
            attr3 = 0 : 0.0  -- 10
            attr3 = 1 : 0.0  -- 13
        attr2 = 1 :
            attr3 = 0 : 0.0  -- 16
            attr3 = 1 : 0.0  -- 14

```

Accuracy on Training data = 0.89375

Accuracy on Testing data = 0.8719211822660099

Extra Credit - Learning Curve

Accuracy for 40 training instances on Testing data = 0.7536945812807881

Accuracy for 80 training instances on Testing data = 0.8719211822660099

Accuracy for 120 training instances on Testing data = 0.8719211822660099

Accuracy for 160 training instances on Testing data = 0.7832512315270936

Accuracy for 200 training instances on Testing data = 0.8719211822660099

Accuracy for 240 training instances on Testing data = 0.8719211822660099

Accuracy for 280 training instances on Testing data = 0.7487684729064039

Accuracy for 320 training instances on Testing data = 0.8719211822660099

Accuracy for 360 training instances on Testing data = 0.8719211822660099

Accuracy for 400 training instances on Testing data = 0.8719211822660099

Accuracy for 440 training instances on Testing data = 0.8719211822660099

Accuracy for 480 training instances on Testing data = 0.8719211822660099

Accuracy for 520 training instances on Testing data = 0.8719211822660099

Accuracy for 560 training instances on Testing data = 0.8719211822660099

Accuracy for 600 training instances on Testing data = 0.8719211822660099

Accuracy for 640 training instances on Testing data = 0.8719211822660099

Accuracy for 680 training instances on Testing data = 0.8719211822660099

Accuracy for 720 training instances on Testing data = 0.8719211822660099

Accuracy for 760 training instances on Testing data = 0.8719211822660099

Accuracy for 800 training instances on Testing data = 0.8719211822660099

